Better Memorization, Better Recall: A Lifelong Learning Framework for Remote Sensing Image Scene Classification

Dingqi Ye, Jian Peng, Haifeng Li, *Member, IEEE*, and Lorenzo Bruzzone, *Fellow, IEEE*

***Abstract*— To infer unknown remote sensing scenarios, most existing technologies use a supervised learning paradigm to train deep neural network (DNN) models on closed datasets. This paradigm faces challenges such as highly spatiotemporal variants and ever-changing scale-heterogeneous remote sensing scenarios. Additionally, DNN models cannot scale to new scenarios. Lifelong learning is an effective solution to these problems. Current lifelong learning approaches focus on overcoming the *catastrophic forgetting* issue (i.e., a successive increase in heterogeneous remote sensing scenes causes models to forget historical scenes) while ignoring the *knowledge recall* issue (i.e., how to facilitate the learning of new scenes by recalling historical experiences), which is a significant problem. This article proposes a lifelong learning framework called asymmetric collaborative network (SCN) for lifelong remote sensing image (RSI) classification. This framework consists of two structurally distinct networks: a preserving network (Pres-Net) and a transient network (Trans-Net), which imitate the long- and short-term memory processes in the brain, respectively. Moreover, this framework is based on two synergistic knowledge transfer mechanisms: triple distillation and prior feature fusion. The triple distillation mechanism enables knowledge persistence from Trans-Net to Pres-Net to achieve better memorization; the prior feature fusion mechanism enables knowledge transfer from Pres-Net to Trans-Net to achieve better recall. Experiments on three open datasets demonstrate the effectiveness of SCN for three-, six-, and nine-task-length learning. The idea of asymmetric separation networks and the synergistic strategy proposed in this article are expected to provide new solutions to the translatability of the classification of RSIs in real-world scenarios. The source codes are available at https://github.com/GeoX-Lab/SCN.**

***Index Terms*— Asymmetric collaborative network (SCN), catastrophic forgetting, knowledge recall, lifelong learning, remote sensing image (RSI) scene classification.**

I. Introduction

With the support of deep neural network (DNN) technology, remote sensing image (RSI) scene classification has made remarkable progress in recent years [1]–[5]. In the most widely used paradigms, DNNs train on closed datasets by supervised learning methods and the models are applied to future scenarios. Different issues may result from this process, including: 1) *data openness*, the continual and rapid increase and accumulation of RSIs; 2) *temporal openness*, the periodic or aperiodic changes of ground feature images in RSIs over time; 3) *spatial openness*, the tendency of the same features to show completely different visual features in different places; and 4) *spectral openness*, the variation in spectral measurements due to the phenological change in ground features. Traditional approaches address the above issue either by retraining models on a mixture of historical data and new data or fine-tuning models on new data. The former approach requires considerable computational cost and storage resources and cannot scale to edge computing devices [6]. The latter approach weakens the model's ability to recognize historical categories, which is known as the *catastrophic forgetting* issue [7]–[9], that is, the model's performance on historical tasks sharply degrades when learning new tasks [10], [11].

Different from the traditional paradigm, lifelong learning technology incrementally learns new data while avoiding these problems. Existing approaches to lifelong learning can be organized into three categories: 1) regularization-based approaches, which are based on design rules that constrain model parameters to reduce the impact of parameter coverage on historical tasks [12]–[15]; 2) dynamic-architecture-based approaches, which are represented by progressive neural networks (PNNs) [16], generate extra parameters for new tasks to reduce the impact of interference between parameters [16]–[19]; and 3) memory-replay-based approaches, which construct a model memory system to alleviate catastrophic forgetting by simulating the interaction between the mammalian hippocampus and neocortex [20], are inspired by the complementary learning system (CLS) theory [21] in neuroscience.

Recently, RSI classification based on lifelong learning has received growing attention and some promising results have been obtained [22]–[25]. However, the majority of current studies focus on overcoming the issue of catastrophic forgetting in neural networks, that is, impaired performance on old tasks when learning new knowledge. However, these studies ignore the critical problem of *knowledge recall*. Indeed, it is still challenging to make full use of memorized historical knowledge to facilitate learning new knowledge due to two reasons: 1) *the stability-plasticity dilemma* [26], that is, better memorization of historical knowledge facilitates learning new

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knowledge, where better learning of new knowledge degrades memory and 2) *knowledge entanglement*, that is, the general knowledge shared among tasks and the specific knowledge related to the individual task are entangled, which impairs the reuse of historical knowledge and specific task learning. In particular, the classification of RSIs in specific tasks is driven by the scale characteristic and channel properties. Therefore, it is important to focus more on the general characteristics for scaling to various tasks.

This research proposes a lifelong learning framework, asymmetric collaborative network (SCN), which is characterized by two core properties: independence and asymmetry. These properties aim to alleviate the stability-plasticity dilemma and the knowledge entanglement problem, respectively. Specifically, SCN exploits two separate networks and regularization strategies. For the network architecture, Pres-Net and Trans-Net encode historical knowledge and new knowledge, respectively, to avoid interference that can result in catastrophic forgetting between different types of knowledge. Moreover, the use of triple distillation loss for regularization allows us to capture the semantic similarity between historical and new knowledge in a feature space [27], ensuring better memorization of distilled knowledge (better memorization). By performing feature fusion before embedding old features and new features into the same low-dimensional feature space to learn their transferability across both networks, valuable knowledge from Pre-Net to complement the learning of Trans-Net is recalled (better recall). SCN also introduces asymmetric architectures. Pres-Net uses a simple network to facilitate the memorization of general features for the long-term retention of old tasks. Meanwhile, Trans-Net uses a more complex task-specific network to better learn the specific features for new tasks. In particular, the multiscale module Res2Block [28] and the multichannel attention module Efficient Channel Attention (ECA) [29] are added to Trans-Net to acquire more representative features.

In summary, the major contributions of our work are outlined as follows.

1) A new lifelong learning framework for RSI classification is proposed to better overcome catastrophic forgetting while facilitating the learning of new unknown categories.
2) A new pair of SCNs are proposed to simulate the abstraction mechanism in human long-term memory and the fine learning mechanism in short-term memory by asymmetric and independent network design. The synergy between the two networks has been achieved via a triple distillation mechanism and prior feature fusion mechanism.
3) Experiments on three open datasets show the effectiveness of the proposed method, which is promising for the classification of RSIs in real-world scenarios.

II. Related Work

In this section, we first introduce the current primary research in lifelong learning, which mainly addresses the problem of catastrophic forgetting. Then, we review research on lifelong learning oriented to RSIs.

A. Overcoming Catastrophic Forgetting in Lifelong Learning

Research on lifelong learning has been exploded in recent years. Related studies primarily address the problem of catastrophic forgetting [7]. Based on strategies for overcoming forgetting, these studies can be categorized into three types of approaches: regularization, dynamic structure, and memory replay.

1) Regularization: The core idea of regularization is to design rules for constrained parameters to reduce the impact of parameter coverage on historical tasks. A representative method includes the Learning without Forgetting (LwF) model proposed by Li and Hoiem [12], which distills knowledge from an old network into a new network by the similarity of outputs. Through the idea of knowledge distillation, this model is able to retain the features of historical tasks to a certain extent [30]–[32] while reducing its performance due to additional distillation loss [33]. Another idea is to constrain plasticity by calculating the importance of synapses to alleviate the weight coverage of old tasks caused by learning new tasks, which is representative of the elastic weight consolidation (EWC) [13] and intelligent synapse (IS) methods [14]. However, the computational complexity based on the Fisher matrix is too high. Pan *et al.* [34] proposed using a Gaussian process formula to obtain higher efficiency of the regularization process. Wang *et al.* [29] proposed Adam-Null Space of Feature Covariance (NSCL) to update the candidate parameters to the null space of all previous tasks through two constraints on stability and plasticity to achieve continual learning. Another idea to mitigate the task-to-task interference by the model structure is in use. Adaptive Random Path Selection Net (Adaptive RPS-Net) [35] encourages the sharing of parameters between tasks through random path selection and achieves the purpose of unlocking the mutual influence between them. Although regularization-based methods are effective in mitigating catastrophic forgetting to a certain extent, they require a tradeoff between old and new tasks due to the inclusion of an additional loss term [36]. In addition, these methods are still prone to forgetting in long-term-to-lifelong learning [37].

2) Dynamic Architecture: This approach preserves task-specific parameters or modules to avoid knowledge interference between old and new tasks. One of the most representative methods is the PNN [16], in which each task is associated with a corresponding network and adapted by lateral links. PNN effectively alleviates the catastrophic forgetting issue since it completely avoids interference of task-specific parameters by separate subnetworks. However, its computational complexity grows exponentially with an increase in the number of tasks, making it difficult to deploy. Some other methods considered the problem of the dynamic construction of the structure. Xiao *et al.* [38] proposed a model that grows in a self-organized manner based on similarity, which enables it to learn faster while inheriting existing model features. However, this model can

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only grow at the top layer, which impairs training efficiency. Yoon *et al.* [18] proposed a dynamic expansion network (DEN), which expands the network with group sparse regularization. Similar studies in [39] and [40] were also designed to break the upper bound of incremental learning heavy model capacity by a dynamic structure.

3) Memory Replay: This approach considers that the memory of historical tasks is motivated by the primary difficulty in incremental learning, in which various factors limit access to historical data. The representative approach is Deep Generation Replay (DGR) proposed by Shin *et al.* [41], which uses a generative network to generate pseudo-historical images for old tasks. Another solution is to store a subset of the representative samples from the previous task to reduce the data storage. Representative methods are the Gradient Episodic Memory (GEM) algorithm proposed by Lopez-Paz and Ranzato [42] and its variants [43], [44]. Although these methods avoid considerable storage of historical images, they fail to address hard convergence, long training sessions, and high hardware requirements issues. Moreover, in long-term learning, this type of replay network still faces the problem of forgetting, which will further lower the quality of replay and affect the final results. Thus, inspired by CLS theory, some works constructed a memory system to mitigate catastrophic forgetting by simulating the interaction between the mammalian hippocampus and neocortex. This approach was first validated in experiments by Hinton and Plaut [45], where a set of plasticity weights were assigned to memory and a set of rapid change weights were assigned to new knowledge. Furthermore, some recent research complemented this idea by generating data distribution [46], pseudo-feature [47], or learning of model parameters [48], [49].

B. Lifelong Learning in Remote Sensing

Some lifelong learning works for RSIs have been proposed in recent years. In terms of regularization-based strategies, Ammour [25] improved the effect on the old land cover classification task by maximizing the distance among tasks. Bhat *et al.* [24] utilized curriculum learning to design a learning order (i.e., from easy to complex) by computing the similarity between new and old classes to obtain better local optima and avoid forgetting. These methods can alleviate catastrophic forgetting to a certain extent. However, it is challenging to ensure the effectiveness of constraints in long sequence tasks, which would lead to considerable forgetting. In terms of dynamic structure, Lu *et al.* [23] suggested using fewer parameters to overcome forgetting, thus avoiding parameter explosion. With respect to memory replay-based methods, Ammour [25] proposed learning how to reproduce data from previous tasks by analyzing the potential data structure of new tasks to alleviate forgetting. Such a proposal still fails to avoid the problems of difficult training and slow convergence.

In summary, despite some progress achieved in remote sensing-oriented lifelong learning, most of the proposed methods only focus on overcoming the catastrophic forgetting issue. To the best of our knowledge, none of the previous research in RSIs considers the problem knowledge recall issue, limiting the performance of current lifelong learning algorithms for real-world scenarios.

III. Methodology

A. Overview

To avoid interference between historical knowledge and new knowledge, we design a novel lifelong learning framework called SCNs, which mimics the human brain's segregate mechanism of long- and short-term memory. As shown in Fig. 1, SCN is an independent-asymmetric architecture with two subnetworks: the preserving network (Pres-Net) and the transient network (Trans-Net), which encode historical knowledge related to old tasks and current knowledge-related new tasks, respectively. Furthermore, Pres-Net acts as a task-general network ResNet [50] to preserve abstract knowledge. In contrast, Trans-Net, equipped with a multiscale feature extraction module and a multichannel attention feature module, acts as a task-specific network to learn fine-grained tasks. This asymmetric-independent architecture ensures the decoupling of accumulated general knowledge (long-term memory) and task-specific knowledge (short-term) while avoiding the fact that the learning of new tasks corrupts the encoding of historical knowledge.

Furthermore, SCN utilizes two collaborative strategies consisting of the triple distillation mechanism and the prior feature fusion mechanism on the two subnetworks, achieving effective knowledge recall (Pres-Net to Trans-Net) and fusion (Trans-Net to Pres-Net). The former mechanism uses a compound loss term to integrate Trans-Net into Pres-Net by maximizing the semantic similarity in the feature space, achieving knowledge persistence and endowing the network with better memorization ability. The latter allows for integrating the prior features stored in Pres-Net with the features learned by Trans-Net to facilitate the learning of new tasks. To ensure the validity of the prior features and effective knowledge recall from Pres-Net to Trans-Net, we design a recall filter according to our previous work [51]. This filter embeds the prior and learned features into the same low-dimensional feature space to learn the transferability of features.

B. Asymmetric-Independent Design of Dual Subnetworks

As mentioned before, SCN consists of two subnetworks with different structures, Pres-Net and Trans-Net, which are responsible for the memorization of old tasks and the learning of new tasks, respectively.

1) Preserving Network: The core function of Pres-Net is to learn abstract knowledge and thus implement long-term memory. Based on our assumption that more abstract knowledge can be acquired by forcing Pres-Net to learn using a simple network structure, we set Pres-Net as a simple network ResNet-50 composed of ordinary residual blocks [50] [ResBlock, shown in Fig. 2(left)], denoted as

$$F_n^P = \mathrm{conv1}\big(\mathrm{conv3}\big(\mathrm{conv1}\big(F_{n-1}^P\big)\big)\big) \tag{1}$$

where F_n^P denotes the output feature of the nth block in Pres-Net, and conv1 and conv3 denote 1×1 and 3×3 convolution operations, respectively.

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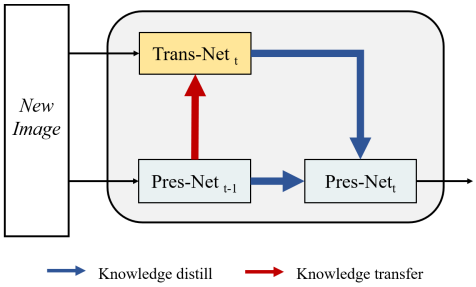


Fig. 1. Incremental learning procedure of the SCN framework. The orange rectangle indicates the Trans-Net learning of task t, and the first cyan rectangle indicates Pres-Net after learning task $t-1$. The learning is divided into two steps. The first step is shown by the red line: the new image is input to Trans-Net and Pres-Net. Trans-Net is learned, while Pres-Net is frozen, and the output features are migrated to Trans-Net to help its learning. The second step is shown by the blue lines, Trans-Net$_t$ and Pres-Net$_{t-1}$ are integrated into a new Pres-Net$_t$ by knowledge distillation, after which Trans-Net will be released. At this point, the whole incremental learning process of the tth task ends.

2) Transient Network: Trans-Net, which is dedicated to learning new tasks, is expected to extract rich semantic information from images. Considering the multiscale and multichannel feature dependence of RSIs, we set up a multiscale feature extraction Res2Block [shown in Fig. 2(right)] and a multichannel attention module ECA (shown in Fig. 3) to help Trans-Net better acquire scale and channel information in new tasks.

a) Multiscale feature extraction module Res2Net: A multiscale feature extraction module Res2Net denoted as $G(a, S)$ is added to Trans-Net. This module can represent multiscale features with a finer degree of regularity by using a larger receptive field. As shown in Fig. 2, a larger receptive field is achieved by the convolution of a set of different scales, denoted as

$$G(a,S) = K(\beta_1, \beta_2, \ldots, \beta_s) \tag{2}$$

$$\beta_i = \begin{cases} \alpha_1, & i = 1 \\ \text{conv3}(\alpha_2), & i = 2 \\ \text{conv3}(K(\alpha_i + \beta_{i-1})), & 2 < i \le s \end{cases} \tag{3}$$

where α denotes the feature after the 1×1 convolution layer, α_i denotes the ith subfeature split by α, $K(\cdot)$ denotes the collocation of features, and S is the scale hyperparameter. The combined convolution first splits the input features into s groups of subfeatures. A convolution operation with kernel size 3×3 is performed on all subfeatures $\alpha_i (i \neq 1)$, after which the result β_i will be combined with the next subfeature α_{i+1} for the next turn of the convolution operation.

In Trans-Net, the output F_T^n of each block is calculated as

$$F_n^T = \text{conv1}\big(G\big(\text{conv1}\big(X_{n-1}^T\big), S\big)\big) \tag{4}$$

where $G(\cdot)$ denotes the multiscale feature extraction module, S denotes the scale hyperparameter, conv1 denotes the 1×1 convolution, and X_{n-1}^T denotes the input feature of the block. It is worth noting that in Trans-Net, the output F_n^T of the nth block is not equal to the input X_n^T of the $(n+1)$th block due to the migration mechanism that we have designed, as detailed in Section III-C.

b) Multichannel attention module ECA: Attention mechanisms improve the capability of networks to focus on critical information in complex contexts. As shown in Fig. 3, a lightweight channel attention module ECA [29] is built in the recall filter to model channel relationships by accessing local cross-channel interaction information among k neighbor channels. As shown in (2), the 1-D convolution is implemented as

$$F_{n-1}^{\prime P} = \text{ECA}\big(F_{n-1}^P\big) = \text{AAP}\big(\text{conv1}\big(F_{n-1}^P\big)\big) \tag{5}$$

where F_{n-1}^P denotes the output features from the nth block of Pres-Net, conv1 denotes the 1×1 convolution, and AAP denotes the adaptive average pooling.

C. Collaboration Between Dual Subnetworks via Transfer Recall Strategy (P→T) and Triple Distill Strategy (T→P)

Two strategies are designed using SCN for collaboration between subnetworks to achieve knowledge recall without catastrophic forgetting: 1) *transfer recall strategy*, which migrates prior knowledge from Pres-Net to Trans-Net by the recall filter and 2) *triple distill strategy*, which integrates the newly learned knowledge from Trans-Net into Pres-Net through triple distill loss.

1) Transfer Recall Strategy (P→T): The catastrophic forgetting and knowledge recall of historical knowledge are considered in SCN lifelong learning. Pres-Net utilizes groups of recall filters [51] to migrate features to Trans-Net, while Trans-Net is learning new categories, that is, before the feature fusion mechanism.

The core idea of the recall filter is to embed the features of both networks into the same low-dimensional feature space, learn the transferability between Pres-Net and Trans-Net features, and finally select the beneficial features for migration, ensuring the validity of migrated features. As shown in Fig. 3, this process contains two parts and is detailed as follows:

$$\begin{aligned} C_{n-1} &= f_{RF}\big(F_{n-1}^{\prime P}, F_{n-1}^P, F_{n-1}^T\big) \\ &= g_{n-1}\big(F_{n-1}^P, F_{n-1}^T\big) \otimes h_{n-1}\big(F_{n-1}^{\prime P}\big) \end{aligned} \tag{6}$$

where $f_{RF}(\cdot)$ denotes the recall filter function, F_{n-1}^P denotes the output of the nth block of Pres-Net, $F_{n-1}^{\prime P}$ denotes the output of F_{n-1}^P after the channel attention module, F_{n-1}^T denotes the output of the nth block of Trans-Net, and $g_{n-1}(\cdot)$ and $h_{n-1}(\cdot)$ represent the embedding and projecting processes in the recall filter, respectively (as shown in Fig. 4).

a) Embedding: The embedding process of the recall filter is represented as $g_{n-1}(\cdot)$. During this process, two linear layers embed features from Pres-Net and Trans-Net into the same space to learn the transferability between them. This process can be described as follows:

$$g_{n-1} = \sigma\big(E_{n-1}^P GAP\big(F_{n-1}^P\big) + E_{n-1}^T \text{GAP}\big(F_{n-1}^T\big) + b\big) \tag{7}$$

where F_{n-1}^P denotes the output features of the $(n-1)$th layer of Pres-Net, g_{n-1} denotes the recall filter, E_{n-1}^P and E_{n-1}^T denote the linear layer outputs of Pres-Net and Trans-Net, respectively, b denotes bias, $\sigma(\cdot)$ is the sigmoid function, and GAP denotes global average pooling.

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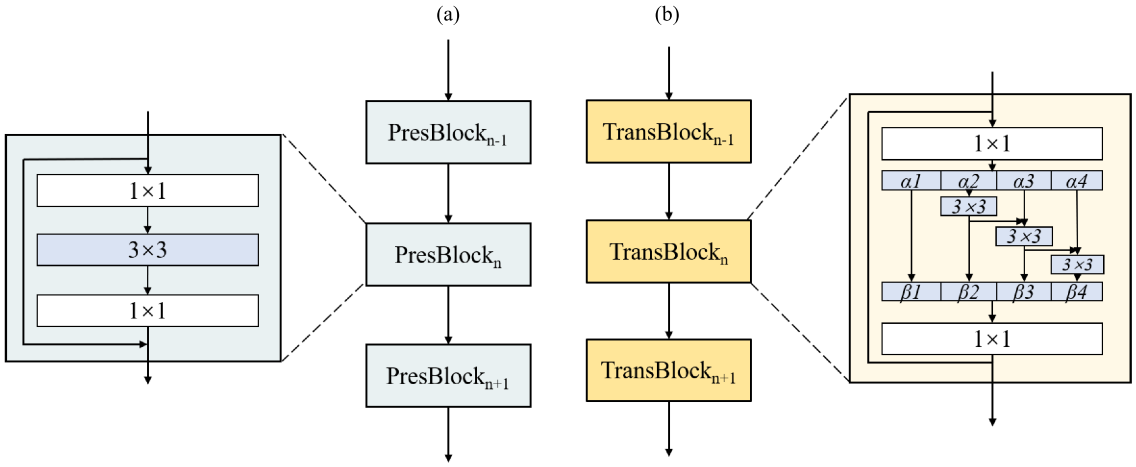


Fig. 2. Structure of (a) Pres-Net and (b) Trans-Net. (Left) The basic structure of Pres-Net, which consists of ordinary residual blocks designed to maintain more historical memory in an abstract and simple structure. (Right) The basic structure of Trans-Net, which replaces the 3 × 3 convolution in Pres-Net with a set of convolutions at different scales to obtain scale information from new images to help learning.

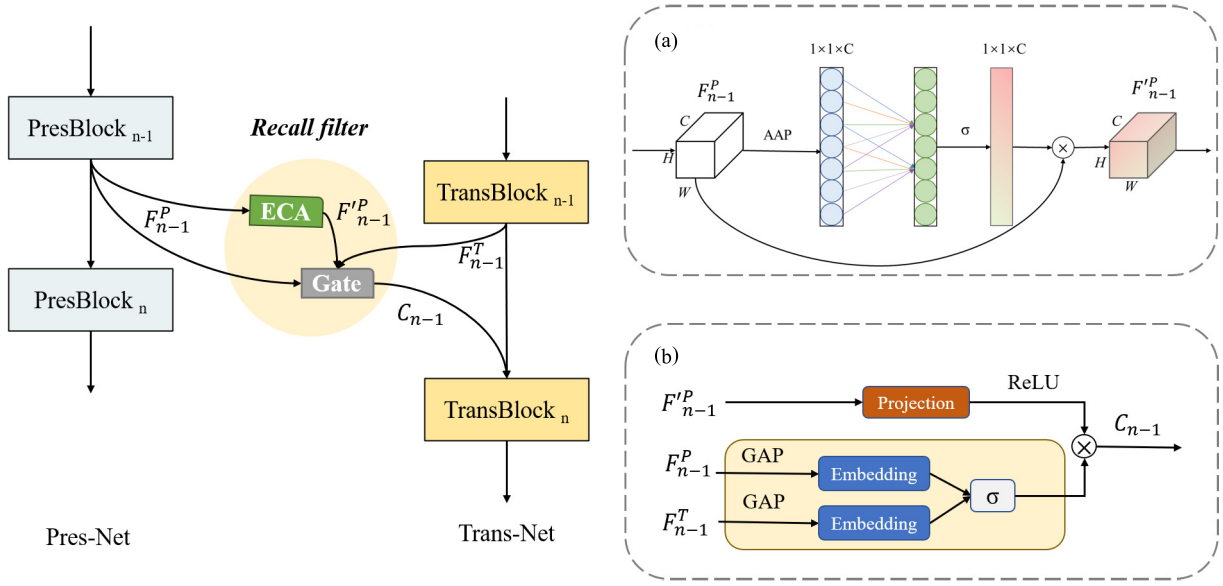


Fig. 3. Knowledge recall from Pres-Net to Trans-Net. Pres-Net and Trans-Net have the same number of layers and communicate between each layer by the recall filter. The recall filter consists of two main modules: a) channel attention module ECA that captures local cross-channel interaction information by considering each channel and its k neighbors and b) transfer gate that learns migrability by embedding Trans-Net (F^T_{n-1}) and Pres-Net (F^P_{n-1}) features into the same space while integrating the projection of features that have passed through the channel attention module (F'^P_{n-1}) and finally integrating them to achieve an effective recall of Trans-Net.

b) Projecting: The projecting process of the recall filter is represented as $h_{n-1}(\cdot)$. The main role of the projecting process is to project features from the channel attention module and achieve alignment. This process can be described as follows:

$$h_{n-1}\left(F'^P_{n-1}\right) = \mathrm{ReLU}\left(P_{n-1} \cdot F'^P_{n-1}\right) \tag{8}$$

where F'^P_{n-1} denotes the output of F^P_{n-1} after the channel attention module, ReLU$(\cdot)$ denotes the rectified linear unit function for activation, and P_{n-1} denotes the projection parameters.

In Trans-Net, the input X^T_n for the nth block is

$$X^T_n = C_{n-1} + F^T_{n-1} \tag{9}$$

where X^T_n denotes the features input of the nth block when training Trans-Net, C_{n-1} denotes the knowledge migrated from Pres-Net, and F_{n-1} denotes the features output by the nth block of Trans-Net.

2) Triple Distill Strategy (T→P): In the second stage of SCN, the new task knowledge preserved by Trans-Net is integrated with the historical knowledge preserved by Pres-Net through a knowledge distillation loss L_{distill} (see Fig. 4). This stage consists of three components

$$L_{\text{distill}} = \gamma\, L_{\text{similar}} + L_{\text{dis_}P} + L_{\text{dis_}T}. \tag{10}$$

L_similar: Semantic similarity generates similar activations in the feature space. For the output of the ith layer of the

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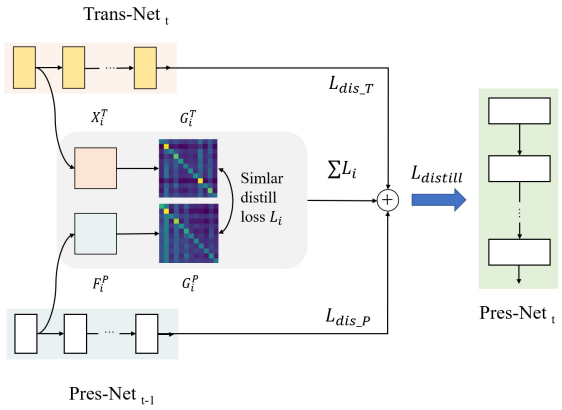


Fig. 4. Triple distillation loss. This loss is composed of three components. For each layer of the output of the two networks, we compute two similarity measures of the features and sum them to obtain L_{similar}. At the same time, we also consider the similarity of the output results of the two networks and calculate $L_{\text{dist_}P}$. In addition, to avoid the extreme case of poor performance of Trans-Net in the transfer recall phase, we calculate $L_{\text{dist_}T}$ to make the data labels as close to the true distribution as possible.

independent network, its activation similarity is defined as

$$G_i = \frac{\hat{X}_i \times \hat{X}_i^T}{||\hat{X}_i \times \hat{X}_i^T||_2} \tag{11}$$

where the output X_i of the ith layer of the network has size $b \times c \times h \times w$, in which b denotes the batch size, c denotes the number of channels, h and w denote the spatial dimensions, X_i^T denotes the deformation of the output X_i of the ith layer of this network, which is a 2-D vector of size $b \times [c \times h \times w]$, $\hat{X}_i \times \hat{X}_i^T$ is a matrix of size $b \times b$, and $||\cdot||$ denotes $L2$ regularization. In Pres-Net and Trans-Net, the similarity loss by G at each layer is defined as

$$L_{\text{similar}} = \frac{1}{b^2} \sum \left\| G_i^T - G_i^P \right\|_F^2 \tag{12}$$

where G_i^T and G_i^P denote the activation similarity of the ith layer of Trans-Net and Pres-Net, respectively, and $||\cdot||_F^2$ denotes the Frobenius norm.

The similarity $L_{\text{dis_}P}$ is also considered in the output distribution of Pres-Net and Trans-Net. $L_{\textbf{dis_}P}$ For Pres-Net, which is dedicated to preserving historical knowledge, the target loss is defined as

$$L_{\text{dis_}P} = L_{\text{CE}}\left(\sum_{i=1} \sigma\left(\frac{X_i^{P'}}{\mathcal{T}}\right), \sum_{i=1} \sigma\left(\frac{X_i^P}{\mathcal{T}}\right)\right) \tag{13}$$

where L_{CE} denotes the cross-entropy loss, P' denotes the updated Pres-Net, $\mathcal{T}$ denotes the distillation hyperparameter temperature, and $\sigma(\cdot)$ denotes the sigmoid function.

$L_{\textbf{dis_}T}$: For Trans-Net, a hybrid loss $L_{\text{dis_}T}$ is used that allows for both data distribution and output distribution to avoid poor learning of new images by Trans-Net. It is defined as follows:

$$\begin{aligned} L_{\text{dis_}T} = {} & (1-\beta) L_{\text{CE}}\left(\sigma\left(X^P\right), Y\right) \\ & + \beta t^2 L_{\text{CE}}\left(\sum_{i=1} \sigma\left(\frac{X_i^{P'}}{\mathcal{T}}\right), \sum_{i=1} \sigma\left(\frac{X_i^T}{\mathcal{T}}\right)\right) \end{aligned} \tag{14}$$

Algorithm 1 Pseudocode for SCN Training and Inference on $N+1$ Tasks

Training:
Input: Input image x and the corresponding label y of the task $N+1$, the weights θ_N^P of Pres-Net that has been trained on previous N tasks, hyper-parameters $\mathcal{T}$,β,$scale$, the number of epochs.
Output: The weights $\theta_{(N+1)}^P$ of Pres-Net which can classify both new and old tasks.
initial Trans-Net weight θ^T
for i = 1 to the number of epochs do:
  Sample $B \in x, y$
  $y = f_T(B, scale, \theta_N^P, \theta^T)$ #Pres-Net transfer to Trans-Net
  Calculate $L = L_{binary_cross_entropy_with_logits}(y, \hat{y})$
  Update θ^T
end for
Extend classifier on Pres-Net (θ_N^P) to generate new model $\theta_{(N+1)}^P$
for j = 1 to the number of epochs do:
  Sample $B \in x, y$
  $\tilde{y} = f_T(B, \theta^T)$; $\bar{y} = f_P(B, \theta_N^P)$
  Calculate $L_{similarity}, L_{dis_T}, L_{dis_P}$ by y,$\tilde{y}$,$\bar{y}$,
  $L = \gamma L_{similarity} + L_{dis_P} + L_{dis_T}$
  Update $\theta_{(N+1)}^P$
End for
Discard Trans-Net

Inference:
Input: Input image x
Output: Output label y that corresponding to image x
  $y = f_p(x)$

where β denotes the hyperparameter to control the effects of the data distribution and output distribution on the final loss and $\mathcal{T}$ is the distillation hyperparameter temperature.

D. Pipeline of Training and Inference for Sequential Tasks

1) Training Phase: As shown in Fig. 5, the training of SCN consists of two stages.

1) **Transfer recall stage:** At the beginning of the stage, Trans-Net is initialized, and Pres-Net is frozen. While Trans-Net learns new tasks, Pres-Net transfers knowledge to Trans-Net through the recall filter (red line).
2) **Triple Distill stage:** At this stage, the knowledge of Trans-Net is integrated into Pres-Net through knowledge distillation. A similarity knowledge distillation loss function, cross similar loss, is invoked to assist in the integration of knowledge (blue line).

2) Inference Phase: As shown in Fig. 5, Trans-Net is removed, and only the Pres-Net remains for inference after completing the training of SCN, which avoids the high computational cost and large parameter size.

The pseudocode for the training procedure is shown in Algorithm 1.

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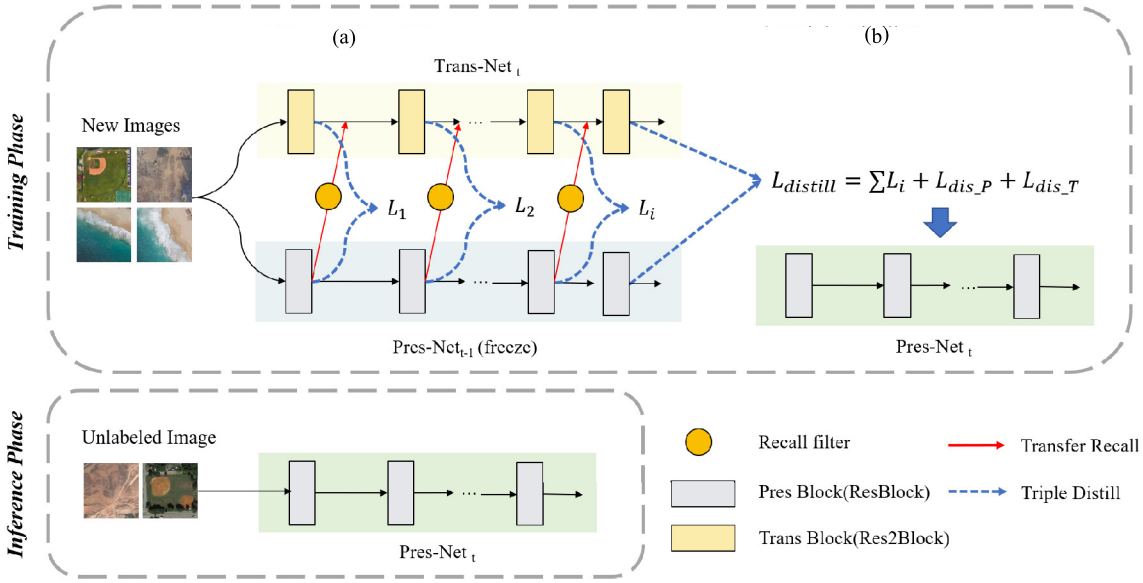


Fig. 5. SCN framework. This schematic shows the whole process of SCN in the training phase and the inference phase. The training phase contains two stages, namely (a) transfer recall stage (red line) and (b) triple distill stage (blue line). The red line indicates the migration of historical experience from Pres-Net to Trans-Net by the recall filter (orange circles) at the transfer recall stage. The blue line indicates the loss of distillation at the triple distill stage, which distills to form Pres-Net$_t$ that retains all current knowledge. In this phase, the parameters of Pres-Net$_{t-1}$ are frozen. The lower part of the figure shows the inference phase of SCN. In this phase, only the Pres-Net$_t$ network, which has saved the previous t tasks, is involved in inference.

IV. Experiments and Results

A. Experimental Setting

We first compare the performance of SCN with some classical methods on class-incremental scenarios using three benchmark datasets for the task of RSI scene classification. Then, we analyze the changes in model capabilities as the number of tasks growing, the impact of the individual modules of SCN, and the robustness of our algorithm to hyperparameters. Finally, we visualize the differences and effectiveness of asymmetric network architectures on feature learning. Specifically, we set three different class-incremental learning scenarios for lifelong learning: three tasks with seven classes for UC-Merced, six tasks with five classes for Aerial Image Dataset (AID), and nine tasks with three classes for Remote Sensing Image Classification Benchmark (RSICB)-256.

1) Datasets: We selected four remote sensing datasets for our experiments. The UC-Merced [52], AID [53], and RSICB-256 [54] datasets were utilized for the comparison of algorithms. Furthermore, the NorthWestern Polytechnical University REmote Sensing Image Scene Classification (NWPU-RESISC45) [55] dataset was utilized for analyzing the effect of the Res2Block module in SCN. The characteristics of these datasets were listed in Table I.

2) Baseline: Six classical lifelong learning methods were chosen for comparison with SCN: **Joint**, **One**, **Finetuning**, **LwF**, **EWC**, and **NAmmours** (2020). Among them, **Joint** and **One** are the ideal-state methods. **Joint** learns all the data together, which avoids catastrophic forgetting while is unable to facilitate new tasks learning. In contrast, **One** learns each task independently, which can avoid catastrophic forgetting while is unable to learn a new task. **Fine-tuning** is a common method that fine-tuning the model corresponding to the previous task to the new task. **LwF** [12] is the most typical function-based approach. **EWC** [13] is one of the most classic regularization-based approaches. The incremental learning framework proposed by **NAmmours** (2020) is a typical regularization approach in remote sensing scene classification, which achieves incremental learning by regular constraints and distillation loss on parameters.

TABLE I
Datasets Description

Datasets	Properties	Characteristics	Experiments
UC Merced	21 categories, 100 images per category, spatial resolution 0.3	Representative dataset, small dataset	IV-B1,IV-B3, IV-B4
AID	30 categories, 220 ~420 images per category, spatial resolution between 0.5 m and 8 m	Representative dataset, inhomogeneous dataset, difficult dataset	IV-B1
NWPU	45 categories, 700 images per category, spatial resolution between 0.2 m and 30 m	Multiscale dataset	IV-B3,IV-B5
RSICB256	35 categories, 800 images per category, spatial resolution between 0.22 m and 3 m	Complex dataset, containing mislabeled noise	IV-B1,IV-B2

3) Metrics: Referring to [56], three metrics were utilized to evaluate the performance of algorithms: *accuracy*, average accuracy (*ACC*), and forward transfer (*FWT*).

1) *Accuracy*: Accuracy represents the ratio of the number of correct model predictions to the total number of predictions, which evaluates the performance of algorithms on the new task, that is,

$$\text{accuracy}(T) = \frac{\text{Num}(\hat{y} = y)}{\text{Num}(y)} \tag{15}$$

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TABLE II
COMPARISON OF ACC ON CLASS-INCREMENTAL LEARNING

Number of sub-tasks	Settings	ACC(%) Joint	One	Finetuning	LwF	EWC	NAmmours	SCN
3 (UC-Merced)	256	**97.14**[2]	**98.3**[1]	45.00	89.29	48.57	43.33	**89.52**[3]
	64	51.95	**92.1**[1]	39.52	**81.19**[3]	35.48	52.86	**82.86**[2]
6 (AID)	256	**96.80**[2]	**96.83**[1]	30.40	88.23	29.73	36.20	**88.93**[3]
	64	48.85	**76.93**[1]	24.85	**75.41**[2]	26.56	43.90	**66.55**[3]
9 (RSICB-256)	256	**99.56**[2]	**99.94**[1]	42.76	81.29	43.69	52.91	**82.63**[3]
	64	**96.46**[2]	**99.02**[1]	41.9	76.39	44.72	50.82	**82.44**[3]

TABLE III
COMPARISON OF FWT ON CLASS-INCREMENTAL LEARNING

Number of sub-tasks	Settings	FWT(%) Joint	One	Finetuning	LwF	EWC	NAmmours	SCN
3 (UC-Merced)	256	\	\	**1.43**[2]	-0.71	-3.57	**2.14**[1]	1.07
	64			**-3.57**[1]	-9.29	-17.14	**-4.64**[2]	**-4.64**[2]
6 (AID)	256	\	\	1.42	0.72	-6.41	**2.11**[1]	**1.55**[2]
	64			-4.20	**-0.18**[2]	-4.39	-4.18	**0.23**[1]
9 (RSICB-256)	256	\	\	-0.35	**0.07**[1]	**0**[2]	-0.14	**0.07**[1]
	64			-1.07	-1.09	-3.39	**0.05**[1]	**0.02**[2]

where T denotes the task, $\hat{y}$ denotes the model output label of the tth task, and y denotes the ground truth label.

2) *ACC* [42]: ACC represents the average accuracy of the model in sequential learning of historical tasks and reflects the model's ability to resist forgetting. It is calculated as follows:

$$\mathrm{ACC}(i) = \frac{1}{T}\sum_{i=1}^{T} P_{T,i} \tag{16}$$

where T denotes the total number of tasks learned by the model, and $P_{T,i}$ denotes the accuracy of the model in learning the ith task.

3) *FWT* [42]: FWT represents the average of the differences between the test accuracy of the model incrementally learning the task and the test accuracy of the model learning the task alone. This metric reflects the influence of the previously learned task on the later learned task and is calculated as follows:

$$\mathrm{FWT}(i) = \frac{1}{T-1}\sum_{i=2}^{T}\left(P_{i,i} - m_i\right) \tag{17}$$

where T denotes the total number of tasks learned by the model, $P_{i,i}$ denotes the accuracy of the ith task after learning the i task, and m_i denotes the accuracy of the ith task through joint learning.

4) Training Details: ResNet-50 was used as the backbone network in our experiments. All baselines were based on the same structure. The training and test sets were divided at a ratio of 4:1. The hyperparameters of the comparison methods were set as follows: λ is 15 in EWC, β is 0.08 in SCN, and $\mathcal{T}$ is 10. The stochastic gradient descent strategy was used in all experiments, with an initialized learning rate in the range $\{0.1, 0.01, 0.0001\}$. The momentum was 0.9, and the weight decay was 1×10^{-5}. The number of training epochs was 100. All methods in the experiments were initialized with kaiming_uniform [57]. The hyperparameter setting of baseline include: LwF $T = 2$, NAmmours $\lambda_1 = 1, \lambda_2 = 1, T = 0.07$, and EWC $\lambda = 50$. The operating environments were Python 3.8 and PyTorch 1.7.1. We used two deep learning stations to run all experiments: DGX A100 Station with 4×80 G GPU and A100 station with 4×40 G GPU.

B. Results and Comparisons

1) Performance Comparison on Class-Incremental Learning: We compared the performance of SCN with the baselines. Tables II and III show the results in terms of ACC and FWT metrics, respectively.

To eliminate the interference of a single dataset, we evaluated the algorithm's lifelong learning ability on three datasets with various characteristics (see Table I). The three-task test used the UC-Merced dataset, while the six-task test used the AID dataset with inhomogeneous data distribution, and the nine-task test used the RSICB-256 dataset with mislabeling noise.

In addition, to reduce the impact of image resolution on incremental learning results, we conducted experiments on all datasets in two learning methods. The first one is 256-size image learning, similar to some common testing processes in the field of remote sensing, where all images were resized to the size of 256×256 (denoted as 256-setting). And the second one is 64-size image learning, where all images were randomly cropped to the size of 64×64 (denoted as 64-setting).

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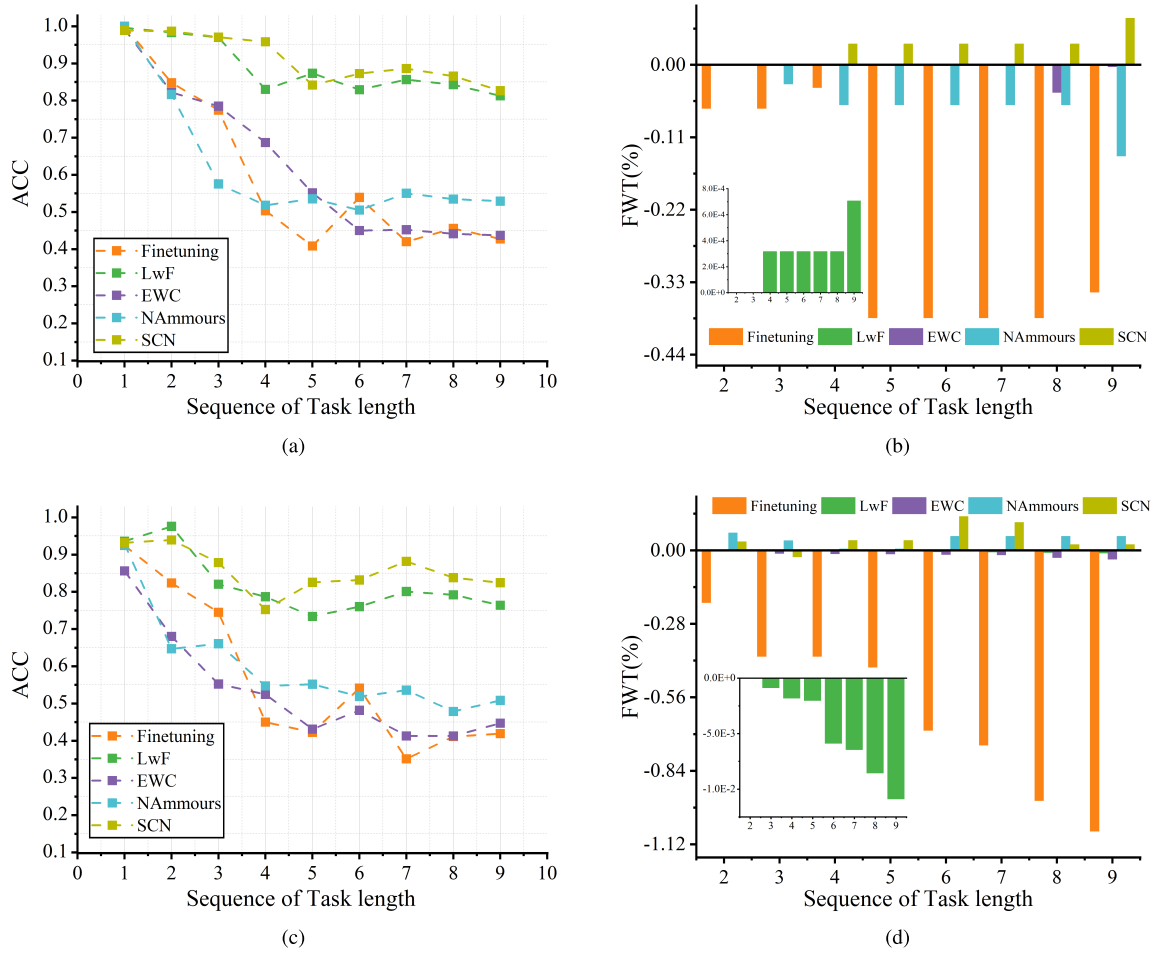


Fig. 6. Performance with sequence length. (a) ACC versus the task sequence length (256-setting). (b) FWT versus the task sequence length (256-setting). The main graph shows the FWT comparison of all methods. Since the range of variation of LwF methods is small, the subplot shows the FWT variation of LwF. (c) ACC versus the task sequence length (64-setting). (d) FWT versus the task sequence length (64-setting). The main graph shows the FWT comparison of all methods. Since the range of variation of LwF methods is small, the subplot shows the FWT variation of LwF.

In the 256-setting, SCN got the first place in ACC among all incremental learning methods (except Joint and One). The incremental learning strategy LwF, which uses a similar distillation method as SCN, also achieved higher ACC score. This indicates that the distillation strategy had better characteristics in incremental learning. However, SCN achieved higher FWT than LwF, which is due to SCN's transfer strategy that benefited the learning of new tasks. Meanwhile, the remote sensing method NAmmours (2020) exhibited a higher score on FWT, while the ACC results were 22.65%∼31.61% lower than SCN. It indicated that its distance constraints at the independent task level and parameter constraints at the shared level are more flexible and have less impact on subsequent tasks.

SCN also shows similar results under 64-setting. SCN preserved the high average accuracy (ACC) after sequential tasks and stays in the top two among similar incremental learning algorithms (except One, Joint), which indicates that SCN has a high ability to overcome catastrophic forgetting. Likewise, the advantage of SCN was more obvious on FWT. Its values reached positive 0.23 and 0.02 in 6- and 9-task sequence learning, and 3-task was also 0%∼12.5% higher compared to other methods. This indicates that SCN is not only resistant to interference from historical tasks in this sequential task learning, but also improves from historical tasks. This is attributed to our knowledge transfer using Pres-Net to Trans-Net in the new task learning, which somewhat generalized and utilized the experience of the historical task.

We noticed that NAmmours (2020) has a gap in overcoming forgetting and obtaining positive migration compared to SCN. In this result, we focused on two points: On the one hand the three-UCMerced did not have the high accuracy as in the original paper NAmmours (2020), which was due to that we are using UC-Merced for three-task learning training and the individual task difficulty was higher than the seven-task training set by NAmmours (2020). On the other hand, we noticed that the ACC of NAmmours (2020) 256-setting is lower than the 64-setting in the three-UC-Merced and six-AID dataset tests. Table IV shows the accuracy of NAmmours (2020) and SCN on each historical task after learning six tasks in sequence. The 256-setting underwent strong forgetting, while the new task is well learned; 64-setting restrained memory, while the new task learning is greatly impaired. This indicates that NAmmours (2020) fails to reach a balance between over-constraint

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TABLE IV
NAMMOURS AND SCN INCREMENTAL LEARNING AFTER SIX TASKS (AID), THE ACCURACY OF EACH TASK

	Settings	1	2	3	4	5	6
NAmmours	256	21.82	23.08	24.68	25.94	23.17	98.53
	64	21.82	30.77	46.84	51.87	51.52	60.59
SCN	256	78.79	72.12	90.51	97.59	96.34	98.24
	64	51.82	69.23	68.99	74.60	65.55	69.12

TABLE V
TASK LEARNING CONTENT OF UC-MERCED

Num	Task content	Number of shadow-prone categories
1	agricultural, **airplane**, baseball diamond, beach, **building**, chaparral, **dense residential**	3
2	forest, freeway, golf course, harbor, intersection, **medium residential**, mobile home park	1
3	**overpass**, parking lot, river, runway, **sparse residential**, **storagetanks**, tennis court	3

and over-relaxation of the old task, while SCN shows more stable performance. This is also reflected in the metrics of the FWT.

It is worth mentioning that One shows the best score on ACC in all three test settings, because it learns each task independently, without sharing of parameters, which means it does not suffer from catastrophic forgetting while is also unable to learn to tasks. We only used it as a reference in an ideal state.

2) Analysis of Sequential Learning on Various Tasks: We further analyzed how the model's learning ability changes as the tasks increase on RSICB-256. Fig. 6 shows the changes in ACC and FWT for each model as the number of tasks increases.

In terms of ACC metrics, SCN maintains ACC at a stable and high level with increasing task volume, both in 256-setting and in 64-setting. In contrast, other methods such as EWC shows a task-volume-related decrease in ACC. It indicates that simple regular methods' limitations in overcoming catastrophic forgetting as tasks increase. Therefore, these models perform limited ability to learn a large number of tasks, which may have fatal consequences for the learning of continuously generated RSIs.

In terms of FWT metric, SCN shows a small improvement in FWT with increasing tasks and obtains positive FWT scores in most cases, which indicates that SCN is more capable of maintaining the learning ability of new tasks in incremental learning compared to the ordinary regular incremental learning methods. It is worth noting that in the case of 256-setting, the FWT scores of methods such as EWC is almost close to 0% in the first few tasks. This is due to the fact that the individual tasks in sequential learning are simple, the model can easily reach an accuracy of 1.0, resulting in the inability to measure the negative migration impact. The regular variation of FWT with task length is more obvious under 64-setting. As shown in Fig. 6, the ACC and FWT score of finetuning, EWC, and LwF decrease as the number of tasks increases. This indicates that the constraint that incremental learning tends to be complex as the number of tasks increases somewhat hinders the learning of new tasks.

3) Ablation Study for Facilitating Better Learning: In this article, we claim that SCN can learn new tasks better without forgetting historical knowledge, primarily due to three modules: 1) ECA, a multichannel feature transfer learning module in the Pres-Net to Trans-Net migration phase; 2) Res2Net, a multiscale feature learning module in Trans-Net; and 3) distillation regularization, an objective function for knowledge distillation in Trans-Net integration into the Pres-Net stage. This section analyzed the impact of different modules for learning a new task separately.

a) Effect of the multichannel attention module ECA: Fig. 7(top) shows the performances of SCN with and without the channel attention module, ECA, with the three-task length setting on the UC-Merced dataset. Fig. 7(a) shows the behavior of ACC during the learning process, and Fig. 7(b) shows the accuracy changes in the coming task. The results show that ECA increases ACC by 5.71% and 0.97% in the first and third tasks, respectively, whereas there is almost no increase in ACC in the second task.

We argue that these results depend on the difference between these three tasks (see Table V). In the first task, there are more scenes, such as airplanes, houses, and dense residential areas, that are prone to shadows. In contrast, relatively few scenes are prone to shadows in the second and third tasks. Note that the addition of the channel attention module ECA improves the anti-interference ability of SCN to cope with problems such as image shadows.

b) Effect of the multiscale feature extraction module Res2Net: Fig. 7(bottom) shows the performance comparison of SCN with and without the multiscale module Res2Net, in the setting of five-task length with the NWPU dataset. NWPU is a remote sensing dataset containing 0.2∼30 m spatial resolution images. Thus, it is appropriate for multiscale feature learning. Fig. 7(c) shows the ACC during the learning process, and Fig. 7(d) shows the accuracy of the model on each task after sequential learning. The results suggest that the accuracy with Res2Net is overall 2.68% higher than the accuracy with ResNet. This indicates that Res2Net to SCN in remote sensing classification task datasets with multiscale features.

c) Effect of similarity-based knowledge distillation loss: As mentioned in Section III, the triple distillation loss in SCN achieves knowledge integration between Trans-Net and Pres-Net. To test its knowledge preservation ability, we compare it with the normal cross-entropy distillation loss and sigmoid loss. Fig. 8 shows the ACC of these three distillation losses in three tasks with seven classes for the UC-Merced dataset. Empirical results show that the ACC associated with triple distillation loss is 0.42% and 1.01% higher than those associated with cross and sigmoid losses, respectively. This suggests that with the triple distillation loss, SCN can better

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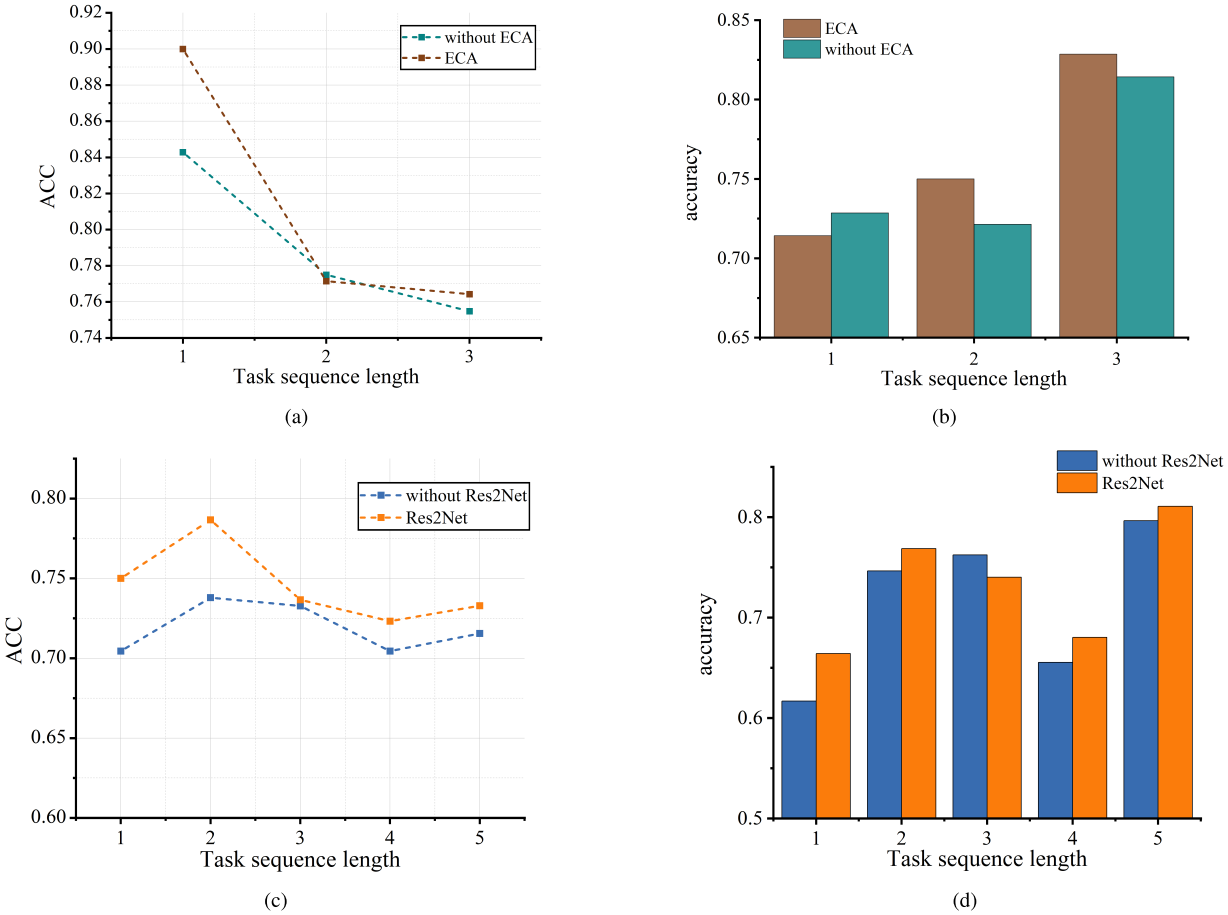


Fig. 7. Feature Extraction Ablation Experiments. The ECA channel ablation experiments were conducted under a three-task UC-Merced experimental setup. The Res2Net module ablation experiments were conducted under a five-task NWPU dataset with a multiscale RSIs experimental setup. (a) ACC and (b) accuracy comparisons including and excluding the ECA module. (c) ACC and (d) accuracy comparison including and excluding the Res2Net module.

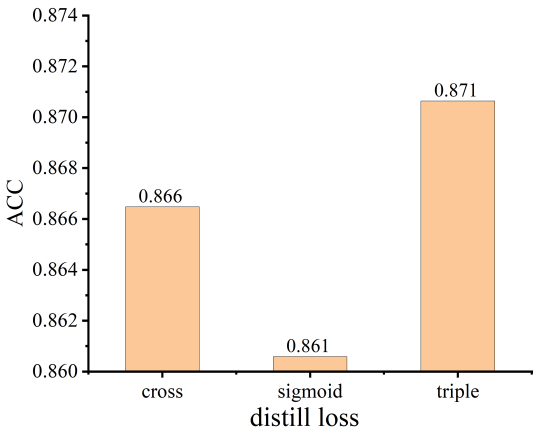


Fig. 8. Knowledge recall from Pres-Net to Trans-Net test under UC-Merced dataset.

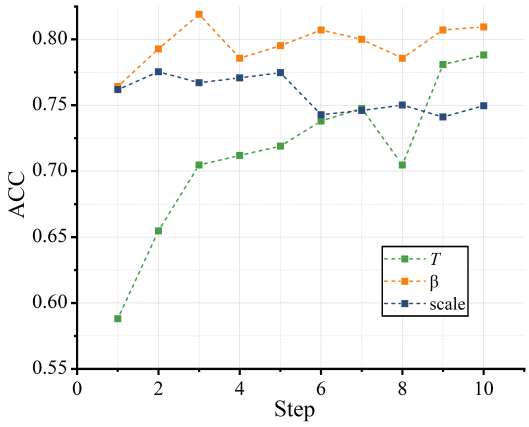


Fig. 9. Behavior of ACC versus the hyperparameter value, where $\mathcal{T}$ step = 1.0, β step = 0.1, and scale step = 1. This test is under UC-Merced dataset.

retain and integrate the knowledge of Trans-Net and Pres-Net to achieve better memorization.

4) Effect of Hyperparameters on the Model: SCN contains three hyperparameters, namely β, $\mathcal{T}$, and scale, where β denotes the weight of the data distribution and S-Net output distribution, $\mathcal{T}$ denotes the temperature in the knowledge distillation stage, and scale denotes the scale in Res2Net.

We analyzed the effect of these hyperparameters on model performance by controlling for single variable changes. Fig. 9 shows the curves of ACC versus the β, $\mathcal{T}$, and scale values: 1) the effect of β on ACC ranges from 0.76 to 0.82, and the

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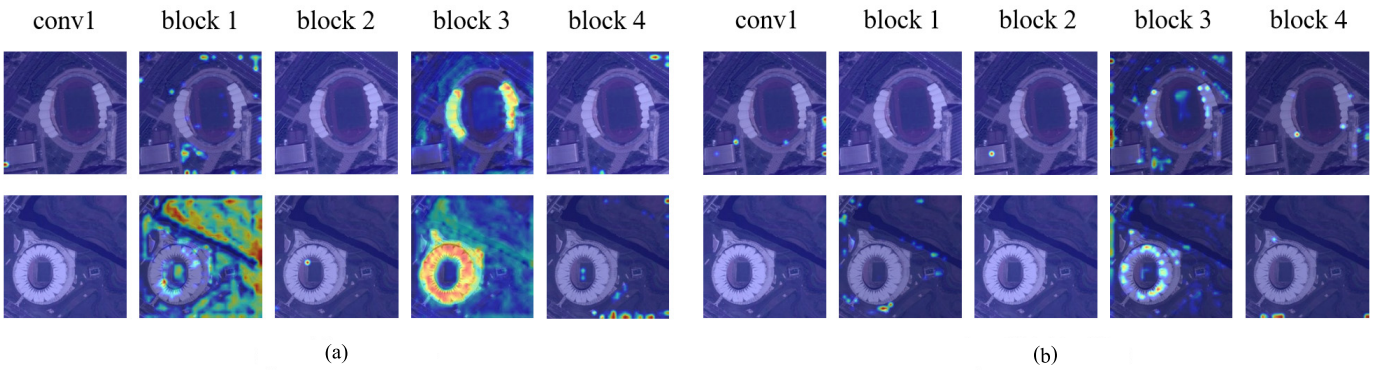


Fig. 10. Grad CAM visualization results using (a) Res2Net and (b) ResNet SCN for stadium scenes (NWPU).

highest value is obtained when β is 0.3; 2) the effect of $\mathcal{T}$ on ACC is more significant and ranges from 0.57 to 0.8, reaching the maximum values when $\mathcal{T}$ is 10; and 3) the effect of scale on ACC on the NWPU dataset is limited, ranging from 0.74 to 0.77. In summary, SCN is more robust to hyperparameters β and scale compared with the hyperparameter $\mathcal{T}$. This also indicates that the value of $\mathcal{T}$ is crucial for the performance of SCN.

5) Feature Visualization Analysis of the Effectiveness of Asymmetric Structures: To verify the effectiveness of the asymmetric design, we visualized SCN with symmetric and asymmetric structures by Gradient-weighted Class ACtivation Mapping (Grad CAM) [58], which is one of the most classic visualization tools. Trans-Net is set up as Res2Net and ResNet to test on the multiscale dataset NWPU. Fig. 10 shows the heatmap of two different scales on the stadium scenes on the final Pres-Net. It shows that the model using Res2Net can locate the information of the overall stadium at different scales more precisely than the model with ResNet. The latter performs relatively worse, focusing on the fragmented parts of the stadium rather than on the stadium itself. It also shows that with the use of Res2Net, the asymmetric design of SCN can be better adapted to the multiscale characteristics of RSIs, hence performing the classification work of remote sensing scenes better.

V. Conclusion

This article proposes SCN, a lifelong learning framework for RSI scene classification, aiming to achieve higher classification accuracy while overcoming catastrophic forgetting. SCN consists of a couple of asymmetric collaborative dual subnetwork functions for learning the new task and memorizing old tasks, called Trans-Net and Pres-Net, respectively. Trans-Net enables better learning of the specific coming task by introducing multiscale and multichannel attention mechanisms into a complex network. At the same time, Pres-Net preserves long-term and general knowledge related to historical tasks via a simpler network. The dual subnetworks cooperate via the triple distillation mechanism, which maximizes the similarity in the feature space and the output distribution to integrate new knowledge of Trans-Net into Pres-Net, achieving effective knowledge integration. Meanwhile, it utilizes the recall filter to learn the transferability between the historical knowledge and new knowledge in networks to integrate prior knowledge from Pres-Net and assists in Trans-Net learning, achieving effective knowledge recall.

The independent network structure design enables SCN to effectively reduce parameter interference during sequential task learning, that is, it alleviates catastrophic forgetting. Furthermore, the knowledge recall mechanism allows SCN to use historical experience to facilitate learning new tasks. In addition, the asymmetric separation of long-term memory networks from short-term memory networks and the synergistic strategy provide a new perspective for RSI scene classification in the real world.

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